# John Cale N. Nombrado

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https://mld-cmd.github.io/John-Cale-Portfolio/ | https://github.com/MLD-cmd

## Personal Summary

I am a Computer Science student seeking a Software Development OJT opportunity, with a strong interest in mobile application development. Experienced in building mobile interfaces and features using Flutter and Dart, supported by academic and personal projects involving APIs, databases, and AI integration. Eager to contribute to a development team, expand my practical skills, and learn from real-world software projects.

## Education

**University of San Jose-Recoletos** — Cebu City, 6000 Cebu
*Bachelor of Science in Computer Science* — Expected Graduation: June 2027

- **Relevant Coursework:** Mobile Development 1,2 and 3, Software Engineering 1 and 2, Data Structures and Algorithm, Information Management 1, Operating Systems, Machine Learning, Applications Development and Emerging Technologies 1 and 2, Automata Theory and Formal languages

## Technical Skills

**Languages:** Python, Dart
**Frameworks & Libraries:** FastAPI, Flutter, OpenCV, Riverpod
**Databases:** PostgreSQL, MySQL
**Developer Tools:** Git, GitHub, MySQL Workbench, Supabase, VS Code, Figma

## Technical Projects

**mAIstra** — *Group Thesis Project* — Jul 2026 – Oct 2026

- Developed a FastAPI OCR service for handwritten C programs using Python, PaddleOCR, OpenCV, and NumPy, including image preprocessing and code cleanup.
- Implemented reading-order logic to associate code continued in a page's right margin or second column, preserving ambiguous results for teacher review.
- Contributed Angular review features including OCR highlighting, multiple-program tabs, extraction-status feedback, and safeguards for unsaved edits.

**Hydro-Tracker** — *Mobile Development* — Jun 2026

- Built a Flutter and Dart hydration app with drink logging, daily goals, intake history, statistics, reminders, and achievements.
- Implemented authentication and cloud data persistence using Supabase, with local caching and editable intake records.
- Integrated weather-based hydration adjustments through Open-Meteo and an AI coach using the Groq API through a Supabase Edge Function.

**OpenEar** — *Software Engineering 1* — Oct – Dec 2025

- Co-developed a Flutter mobile study app for visually impaired learners, with voice-guided access to study notes and quizzes.
- Implemented voice-based questions about uploaded notes and refined the mobile question flow.
- Improved speech recognition, voice navigation, and audio feedback across app screens.

**MeshUnwarp** — *Graphics and Visual Computing* — Jun 2026

- Developed a Python and Tkinter desktop workflow for document image processing, dewarping, cleanup, and text extraction.
- Integrated OpenCV, NumPy, PyTorch, and pretrained DocTr models for image preprocessing and document correction.
- Configured Apple Vision as the primary OCR engine with EasyOCR as a fallback, using background processing to keep the interface responsive.

## Certifications

- **AI Engineer for Developers Associate** – DataCamp (November 2026), Credential ID: AIEDA0012590713793
- **TOPCIT Level 2** (Test of Practical Competency in ICT) – IITP, Score: 244 (July 2026)